

SURAJ KUMAR SAHU

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PhD candidate in Physics studying mechanobiology of cell–cell and cell–matrix interactions, collective cell motility, multicellular patterning, and network function. Develops agent-based, active-gel, and computational models of vascular network formation, cell-junction dynamics, and extracellular-matrix remodeling.

EDUCATION

University of California Merced	Jan 2021 – Present
PhD Candidate, Physics Advisor: Prof. Ajay Gopinathan	
National Institute of Technology Rourkela	Aug 2017 – May 2019
Master of Science in Physics	
D.R. Nayapalli College, Utkal University	Jul 2014 – Jun 2017
Bachelor of Science (Honors in Physics)	

PUBLICATIONS

Sahu, S. and M. Biswas. “Modeling protein association from homogeneous to mixed environments: A reaction-diffusion dynamics approach.” *Journal of Molecular Graphics and Modeling* 107, 107936 (2021).

RESEARCH EXPERIENCE

Graduate Research Assistant Gopinathan Group, UC Merced	Jan 2021 – Present
- Developed a computational model of collagen-matrix compaction and remodeling driven by contractile forces in multicellular fibroblast networks, in collaboration with Dasbiswas Lab. <i>In preparation</i>. - Developed an active-gel model coupling actomyosin-cortex mechanics, cadherin clustering, and external loading to analyze cell–cell contact formation, junction maturation, and rupture. <i>In preparation</i>. - Developed an agent-based model of vascular network formation and remodeling from cell migration and multicellular interactions; characterized network topology and transport function in collaboration with Sindi Lab and McCloskey Lab.	
Graduate Teaching Assistant Department of Physics, UC Merced	Jan 2021 – Present
- Led experimental labs and discussions for introductory physics courses; served as teaching assistant for upper-division course. - Graduate Teaching Assistant for PHYS 115, PHYS 009, PHYS 009L, PHYS 018, PHYS 008L across multiple terms (2021–2026).	
Graduate Student Researcher Computational Biophysics Group, NIT Rourkela	Aug 2017 – May 2019
Coarse-grained reaction-diffusion (ReaDDy) modeling of macromolecular crowding effects on protein association; showed crowder size dominates over shape in stabilizing products; characterized additivity rules in mixed-crowder systems. Published in <i>J. Mol. Graph. Model.</i> (2021).	

TECHNICAL EXPERTISE

- Modeling & Scientific Computing:** Agent-Based Modeling, Active-Gel Modeling, Numerical Simulations, Particle-Based Simulations, Reaction-Diffusion Systems, Network Dynamics, HPC Workload Management and Batch-Job Scheduling with Slurm
- Data & Research Software:** Data Analysis and Modeling, Scientific Visualization, Image Analysis, Python, Julia, NumPy, SciPy, NetworkX, ReaDDy, LaTeX, Git, GitHub, VS Code
- AI-Enabled Research:** Agentic Scientific Workflow Design, AI-Assisted Computational Reproducibility Audits, Context Engineering, Model Context Protocol (MCP) Tool Development (**Cellpose-MCP**), AI-Enabled Image Analysis
- LLM Research Tools:** Claude, Codex

PRESENTATIONS & WORKSHOPS

Organizer & Instructor

- **Spring 2025, UC Merced:** Organized and taught a graduate workshop on AI tools for research and data analysis, covering context engineering, agentic scientific workflows, data and image analysis, and research automation.
- **Summer 2024, University of Pennsylvania:** Led a research project and instructed graduate trainees during the **Center for Engineering and Mechanobiology annual boot camp**.

Selected Oral Presentations:

- **BPS 2026**, San Francisco – “Collective Cell Motility of Fibroblasts Driven by Contractile Multicellular Network Formation”
- **APS March Meeting 2025**, Los Angeles – “Stability of Cell-Cell Junctions: Balancing Cortical Tension and Cadherin Aggregation”
- **Cell Bio 2024, ASCB/EMBO**, San Diego – “Balancing Cortical Tension and Adhesive Force for Stable Cell Junctions”
- **APS March Meeting 2024**, Minneapolis – “Modeling Cell-Cell Junction Mechanics in Vascular Networks”
- **APS March Meeting 2023**, Las Vegas – “Assembly and Mechanical Remodeling of Vascular Network”
- **APS March Meeting 2022**, Chicago – “Agent-Based Simulation of Vasculogenesis”

MENTORING, AWARDS & LEADERSHIP

Graduate Student Research Mentor | I-BEST NSF REU, UC Merced

Jun – Aug 2026

- Mentored **Akshay Langhani**, a first-year Illinois State University undergraduate, on “Quantifying Biological Functionality of In Silico Vascular Networks.”

Awards & Achievements:

- **2025:** CEMB Summer Research Fellowship; CCBM Travel Award; Physics Graduate Group Travel Fellowship
- **2024:** GradExcel Peer Mentor Award; Physics Graduate Group Travel Fellowship
- **2023:** CCBM Outreach Fellowship; CCBM Travel Fellowship; Physics Graduate Group Travel Fellowship
- **2022:** Bobcat Summer STEM Academy Fellowship; Physics Graduate Group Travel Fellowship

Trainee Leadership Council | Center for Engineering and Mechanobiology (CEMB)

Aug 2023 – Aug 2025

- Planned and organized tutorials, research presentations, and professional development workshops for CEMB graduate trainees across multiple institutions.

President | Graduate Biophysics Club, UC Merced

Jun 2021 – Jun 2023

- Led biophysics journal club, organized science outreach events and workshops, and coordinated professional development and networking opportunities for graduate students.

GradExcel Peer Mentor | UC Merced Graduate Division

Aug 2024 – Aug 2025

- Mentored incoming graduate students on personal well-being, academic success, and professional development; recognized with GradExcel Peer Mentor Award (2024).

REFERENCES

Prof. Ajay Gopinathan ✉

Department of Physics, CCBM
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University of California Merced
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Prof. Suzanne Sindi ✉

Department of Applied Mathematics
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Assoc. Prof. Kinjal Dasbiswas ✉

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